Light Availability Strongly Alters Belowground Biomass in C_3 and C_4 Grasses

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BACKGROUND

- Plants respond to nutrient addition and light availability by increasing biomass.
- Plants preferentially allocate biomass aboveground in response to increasing nutrient availability and belowground in response to increasing light availability.
- Understanding biomass allocation responses to interactions between nutrient addition and light availability remains under-examined, particularly in reference to phosphorus availability and between photosynthetic pathways.

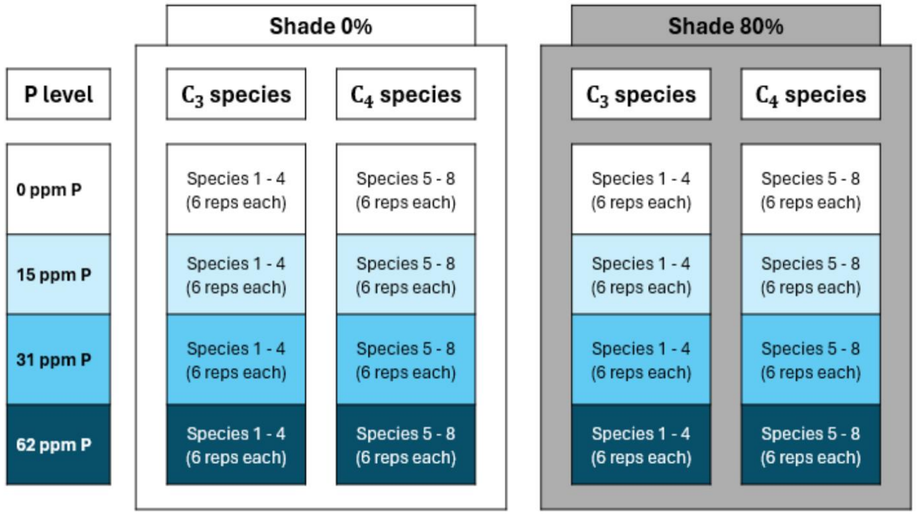
C_3 grasses

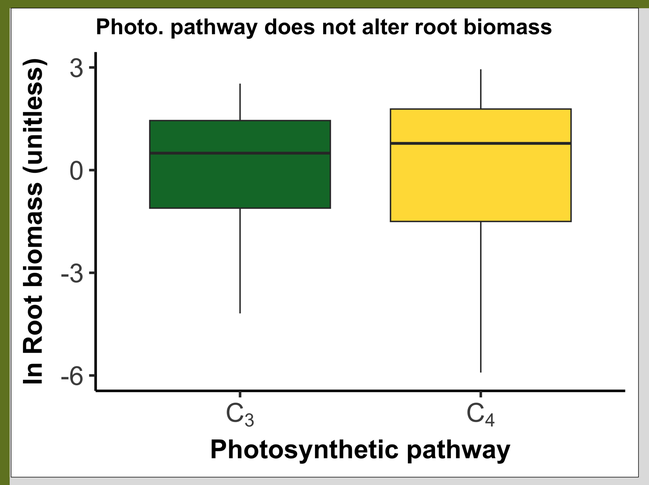
- *Dichanthelium oligosanthes*
- *Dichanthelium scoparium*
- *Dichanthelium clandestinum*
- *Panicum antidotale*

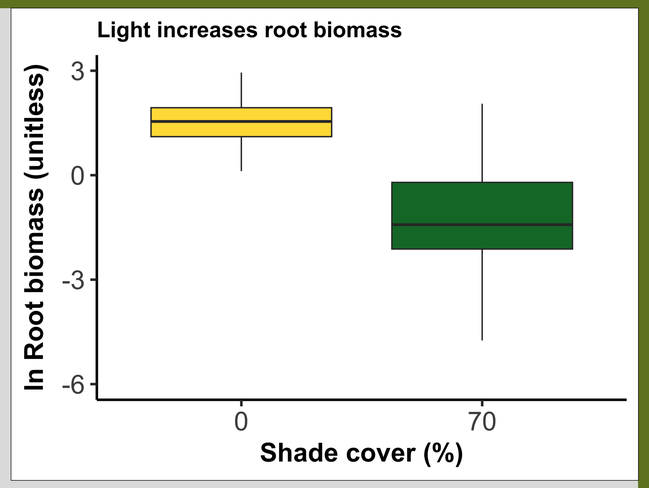
C_4 grasses

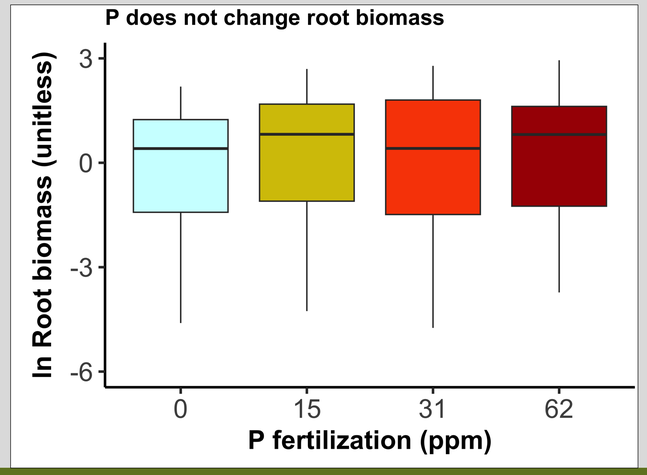
- *Panicum antidotale*
- *Panicum trichanthum*
- *Panicum coloratum*
- *Panicum hallii*

METHODS









RESULTS/CONCLUSION

- While root biomass did not differ between C_3 and C_4 species or across the P fertilization gradient, root biomass generally increased with increasing light availability (i.e., decreasing shade cover).
- Root biomass increased more strongly with increasing light availability in C_4 species than C_3 species.
- These findings could indicate that enhanced demand to build and maintain photosynthetic enzymes in C_4 species may drive increased belowground biomass allocation as a strategy for increasing nutrient and water acquisition.
- Ongoing work is centered around quantifying aboveground biomass responses to these treatment combinations, which will allow us to address questions related to biomass partitioning between above- and belowground biomass.